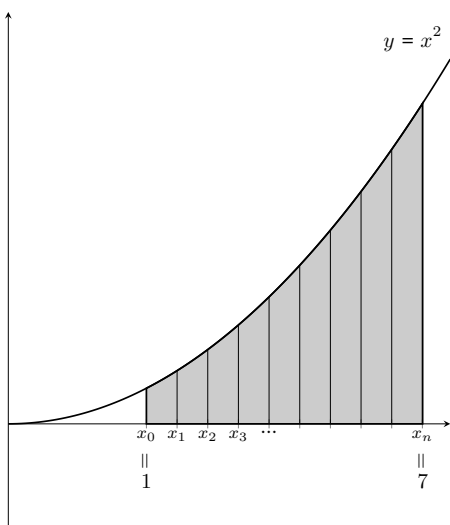


Definition and Properties of the Definite Integral

Today's Goals

- ◆ You should be able to write a Riemann sum that approximates a definite integral.
- ◆ You should be able to state the definition of the definite integral as a limit of Riemann sums and recognize how this definition could be applied to compute a definite integral.
- ◆ You should be able to apply the basic properties of the definite integral to simplify expressions involving definite integrals.

- Hyacinth is interested in finding the area under $y = x^2$ between $x = 1$ and $x = 7$. She decides that she'll first estimate this area using a right-hand sum with n rectangles. So, she slices the area into n slices of equal width Δx . She labels the endpoints of the slices $x_0 = 1, \dots, x_n = 7$ as shown in the picture below.



- Express Δx in terms of n .

Solution

Since the interval $[1, 7]$ is 6 units wide, if we're going to use n rectangles, the width of each rectangle must be $\frac{6}{n}$.

- Express x_1 , x_2 , and x_{12} in terms of Δx .

Solution

x_1 is one rectangle width to the right of $x_0 = 1$, so $x_1 = 1 + \Delta x$. Similarly, x_2 is 2 rectangle widths to the right of $x_0 = 1$, so $x_2 = 1 + 2\Delta x$. By the same reasoning, $x_{12} = 1 + 12\Delta x$.

- Generalize: express x_k in terms of k and Δx ; your formula should work for $k = 0, \dots, n$.

Solution

In general, $x_k = 1 + k\Delta x$.

- (d) Remember that Hyacinth has decided to approximate the area using a right-hand sum. What is the area of the first approximating rectangle? What is the area of the second approximating rectangle? (Express your answers in terms of $x_0, x_1, x_2, \dots, x_n$ and Δx .)

Solution

The first rectangle has width Δx and height x_1^2 , so its area is $x_1^2 \Delta x$. Similarly, the area of the second rectangle is $x_2^2 \Delta x$.

- (e) Write out R_n , the right-hand sum with n rectangles that approximates the entire area from $x = 1$ to $x = 7$. Your sum should still be in terms of $x_0, x_1, \dots, x_n$ and Δx .

Solution

We simply want to add up the areas of all of the rectangles, which is

$$x_1^2 \Delta x + x_2^2 \Delta x + \cdots + x_n^2 \Delta x.$$

We can write this in sigma notation as $\sum_{k=1}^n x_k^2 \Delta x$.

Note: We don't have a way to evaluate this by hand, but [Wolfram Alpha](#) can evaluate it. To have Wolfram Alpha evaluate it, we first need to plug in the meaning of x_k and Δx :

$$\begin{aligned} R_n &= \sum_{k=1}^n x_k^2 \Delta x \\ &= \sum_{k=1}^n (1 + k\Delta x)^2 \Delta x \quad (\text{plugging in } x_k = 1 + k\Delta x) \\ &= \sum_{k=1}^n \left(1 + k \cdot \frac{6}{n}\right)^2 \frac{6}{n} \quad (\text{plugging in } \Delta x = \frac{6}{n}) \end{aligned}$$

We can tell Wolfram Alpha to evaluate this by typing

`sum (1 + k * 6/n)^2 * 6/n, k from 1 to n`

(Wolfram Alpha says that the sum is equal to $\frac{6(19n^2 + 24n + 6)}{n^2}$.)

- (f) Write a limit that gives the *exact* area Hyacinth is looking for.

Solution

We get better approximations by using more rectangles, so the exact area $\int_1^7 x^2 dx$ is equal to $\lim_{n \rightarrow \infty} R_n$.

Note: Since Wolfram Alpha told us that $R_n = \frac{6(19n^2 + 24n + 6)}{n^2}$, this limit is equal to $\lim_{n \rightarrow \infty} \frac{6(19n^2 + 24n + 6)}{n^2}$, which is 114.

(g) Write out L_n in terms of $x_0, x_1, \dots, x_n$ and Δx .

Solution

If we're doing a left-hand sum with n rectangles, the area of the first rectangle is $x_0^2 \Delta x$, the area of the second rectangle is $x_1^2 \Delta x$, and so on. So, the total area in the rectangles is

$$\boxed{x_0^2 \Delta x + x_1^2 \Delta x + \cdots + x_{n-1}^2 \Delta x}, \text{ which can be written in sigma notation as } \boxed{\sum_{k=0}^{n-1} x_k^2 \Delta x}.$$

The Definition of the Definite Integral

The definite integral of $f(x)$ from $x = a$ to $x = b$, denoted $\int_a^b f(x) dx$, is defined to be

$$\lim_{n \rightarrow \infty} \underbrace{\sum_{k=1}^n f(x_k) \Delta x}_{R_n} = \lim_{n \rightarrow \infty} \underbrace{\sum_{k=0}^{n-1} f(x_k) \Delta x}_{L_n}, \text{ where } \Delta x = \frac{b-a}{n} \text{ and } x_k = a + k\Delta x$$

provided these limits exist. (If f is continuous on $[a, b]$, then the limits always exist.)

Note on notation: The variable x in $\int_a^b f(x) dx$ is a “dummy variable”: $\int_a^b f(x) dx$, $\int_a^b f(t) dt$, and $\int_a^b f(\clubsuit) d\clubsuit$ all mean the same thing.

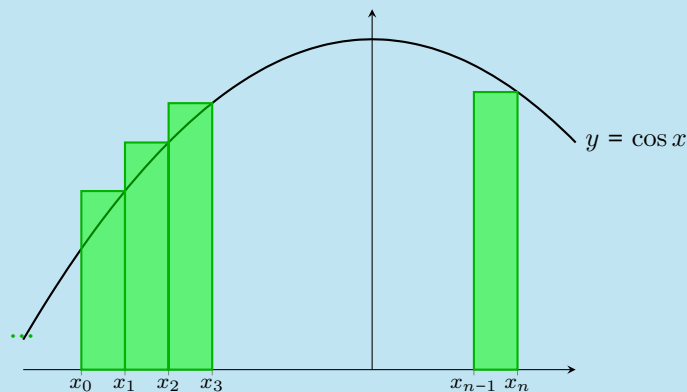
2. Bob is looking at some function $f(x)$ and estimating an area under $y = f(x)$ using a right-hand sum. The sum he's written is

$$R_n = \sum_{k=1}^n (\cos x_k) \Delta x \text{ where } \Delta x = \frac{1}{2n} \text{ and } x_k = -\frac{1}{3} + k\Delta x.$$

What area is Bob estimating?

Solution

The expression $\sum_{k=1}^n (\cos x_k) \Delta x$ can be written out as $(\cos x_1) \Delta x + (\cos x_2) \Delta x + (\cos x_3) \Delta x + \cdots + (\cos x_n) \Delta x$; we can visualize it as the area of rectangles like this (we've shown only the first few and the last rectangle, but there should be n rectangles in all), where the width of each rectangle is Δx :



So, this is an approximation for the area under $\cos x$ between $x = x_0$ and $x = x_n$. The formulas $\Delta x = \frac{1}{2n}$ and $x_k = -\frac{1}{3} + k\Delta x$ say that

$$x_0 = -\frac{1}{3} + 0\Delta x = -\frac{1}{3}$$

$$x_n = -\frac{1}{3} + n\Delta x = -\frac{1}{3} + n \cdot \frac{1}{2n} = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$$

So, Bob is estimating the area under $y = \cos x$ between $x = -\frac{1}{3}$ and $x = \frac{1}{6}$.

3. Let $f(x) = \cos x$. Suppose we want to approximate $\int_0^{1.5} f(x) dx$ using various left-hand and right-hand sums.

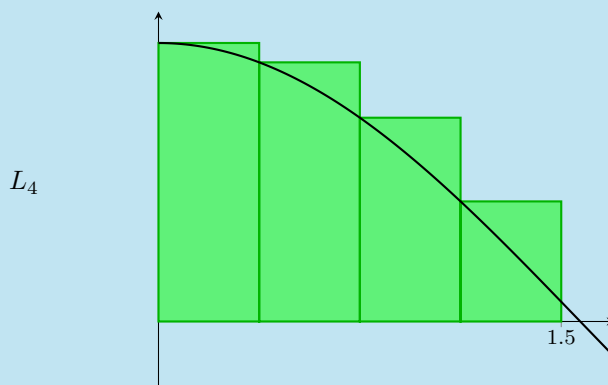
(a) Put the following expressions in ascending order.

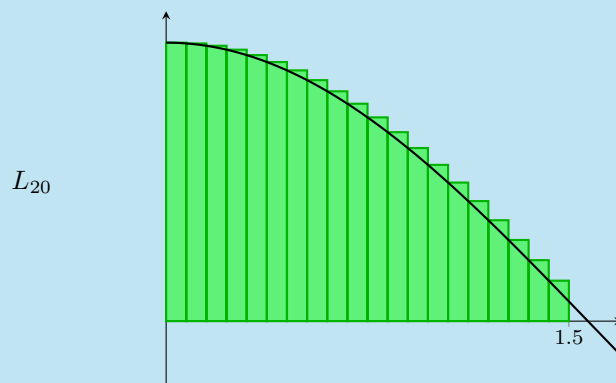
$$\int_0^{1.5} f(x) dx \quad L_4 \quad R_4 \quad L_{20} \quad R_{20} \quad L_{100} \quad R_{100}$$

Solution

Since $f(x) = \cos x$ is decreasing on $[0, 1.5]$, left-hand sums will be overestimates for $\int_0^{1.5} f(x) dx$, and right-hand sums will be underestimates.

Here are pictures of L_4 and L_{20} :





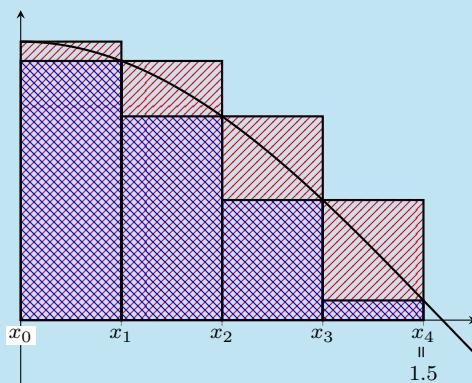
As we can see, L_{20} is much closer to $\int_0^{1.5} f(x) dx$ than L_4 is. By the same idea, L_{100} will be even closer to $\int_0^{1.5} f(x) dx$ than L_{20} is. Similarly, among the right-hand sums, R_{100} is closest to $\int_0^{1.5} f(x) dx$, followed by R_{20} and then finally by R_4 . That gives us the following ordering:

$$R_4 < R_{20} < R_{100} < \int_0^{1.5} f(x) dx < L_{100} < L_{20} < L_4.$$

- (b) Find $|R_4 - L_4|$. How can you visualize this? (What's the least work you can do to find $|R_4 - L_4|$? Do you need to calculate R_4 and L_4 ?)

Solution

Here are pictures of L_4 (in red stripes) and R_4 (in blue stripes):



Notice that the first 3 rectangles in R_4 are the same as the last 3 rectangles in L_4 ; that is,

$$R_4 - (\text{last rectangle in } R_4) = L_4 - (\text{first rectangle in } L_4).$$

Rearranging this equation,

$$\begin{aligned} R_4 - L_4 &= (\text{last rectangle in } R_4) - (\text{first rectangle in } L_4) \\ &= f(1.5)\Delta x - f(0)\Delta x \text{ where } \Delta x = \frac{1.5}{4} \text{ is the width of each rectangle} \end{aligned}$$

So,

$$\begin{aligned}|R_4 - L_4| &= |f(1.5) - f(0)|\Delta x \\ &= |\cos 1.5 - 1| \frac{1.5}{4}\end{aligned}$$

Since $\cos 1.5 < 1$, $\cos 1.5 - 1$ is negative, so $|\cos 1.5 - 1| = -(\cos 1.5 - 1) = 1 - \cos 1.5$:

$$= \boxed{(1 - \cos 1.5) \frac{1.5}{4}}$$

Alternate Method: If you prefer, you could write out both L_4 and R_4 :

$$\begin{aligned}R_4 &= f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x + f(x_4)\Delta x \\ L_4 &= f(x_0)\Delta x + f(x_1)\Delta x + f(x_2)\Delta x + f(x_3)\Delta x\end{aligned}$$

Subtracting, $R_4 - L_4 = f(x_4)\Delta x - f(x_0)\Delta x = f(1.5)\Delta x - f(0)\Delta x$, as we said before.

- (c) Find $|R_{100} - L_{100}|$.

Solution

By the same reasoning as in (b), $|R_{100} - L_{100}| = |(\cos 1.5)\Delta x - (\cos 0)\Delta x|$ where now $\Delta x = \frac{1.5}{100}$ (it's the width of each rectangle if we use 100 rectangles). This can be rewritten as

$$\boxed{(1 - \cos 1.5) \frac{1.5}{100}}.$$

- (d) How large do we need n to be ensure $|R_n - L_n| < 0.05$? Why might we care to have $|R_n - L_n| < 0.05$?

Solution

Using the same reasoning as in (b) and (c), $|R_n - L_n| = (1 - \cos 1.5) \frac{1.5}{n}$. We want n to be large enough to ensure that this is < 0.05 , so we want $(1 - \cos 1.5) \frac{1.5}{n} < 0.05$. Rearranging this, we need $n > (1 - \cos 1.5) \frac{1.5}{0.05} = (1 - \cos 1.5)30$.

Since 1.5 is just a little bit less than $\frac{\pi}{2}$, $\cos 1.5$ is just a bit bigger than 0, so $1 - \cos 1.5$ is just a bit less than 1. Therefore, if we take n to be at least 30, we can be sure that $n > (1 - \cos 1.5)30$.

As we said in (a), left-hand sums are overestimates for $\int_0^{1.5} \cos x \, dx$ and right-hand sums are underestimates, so we always have $L_n > \int_0^{1.5} \cos x \, dx > R_n$. If R_n and L_n are within 0.05 of each other, that means we have $\int_0^{1.5} \cos x \, dx$ pinned within two values that are within 0.05 of each other. This means that we can estimate $\int_0^{1.5} \cos x \, dx$ and be certain that we are off by less than 0.05.

4. (a) How do $\int_1^4 f(x) \, dx$ and $\int_4^1 f(x) \, dx$ relate? Explain.

Solution

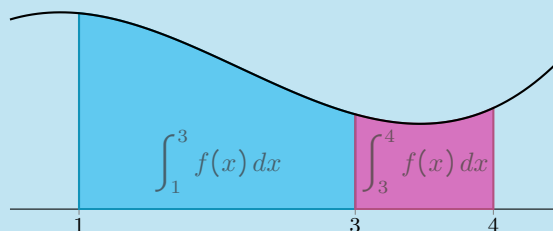
They are negatives of each other; that is, $\int_4^1 f(x) dx = -\int_1^4 f(x) dx$. This makes sense if you think about the net change interpretation: if $f(x)$ is the rate at which water is flowing into / out of a pool at time x , then $\int_1^4 f(x) dx = (\text{amount of water in pool at time 4}) - (\text{amount of water in pool at time 1})$, while $\int_4^1 f(x) dx = (\text{amount of water in pool at time 1}) - (\text{amount of water in pool at time 4})$.

This idea generalizes: $\int_b^a f(x) dx = -\int_a^b f(x) dx$ for any a and b .

- (b) Can you simplify $\int_1^3 f(x) dx + \int_3^4 f(x) dx$? How about $\int_1^3 f(x) dx + \int_3^2 f(x) dx$?

Solution

From a picture, we can see that $\int_1^3 f(x) dx + \int_3^4 f(x) dx = \int_1^4 f(x) dx$:



On the other hand,

$$\begin{aligned} \int_1^3 f(x) dx + \int_3^2 f(x) dx &= \int_1^3 f(x) dx - \int_2^3 f(x) dx \text{ by the result of (a)} \\ &= \int_1^2 f(x) dx \end{aligned}$$

So, in general, $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$.

- (c) How does $\int_a^b 3f(x) dx$ relate to $\int_a^b f(x) dx$? Why?

Solution

$\int_a^b 3f(x) dx = 3 \int_a^b f(x) dx$. One way to think about this is to picture it in terms of signed area: the graph of $3f(x)$ is just the graph of $f(x)$ stretched vertically by a factor of 3 (to be 3 times as tall), so the area under $3f(x)$ from $x = a$ to $x = b$ is 3 times the area under $f(x)$ from $x = a$ to $x = b$.

- (d) How does $\int_a^b [f(x) + g(x)] dx$ relate to $\int_a^b f(x) dx$ and $\int_a^b g(x) dx$? Why?

Solution

$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$. One way to understand this is by thinking about net change. For example, imagine that a swimming pool is being filled by two pipes, one filling at the rate $f(x)$ and the other at the rate $g(x)$. Then, the total rate at which water is being added to the pool at time x is $f(x) + g(x)$, and

$$\underbrace{\left(\begin{array}{c} \text{amount of water} \\ \text{added between} \\ \text{time } a \text{ and time } b \end{array} \right)}_{\int_a^b [f(x) + g(x)] dx} = \underbrace{\left(\begin{array}{c} \text{amount of water added} \\ \text{by pipe 1 between} \\ \text{time } a \text{ and time } b \end{array} \right)}_{\int_a^b f(x) dx} + \underbrace{\left(\begin{array}{c} \text{amount of water added} \\ \text{by pipe 2 between} \\ \text{time } a \text{ and time } b \end{array} \right)}_{\int_a^b g(x) dx}$$

Properties of the Definite Integral

- ◆ $\int_b^a f(x) dx = - \int_a^b f(x) dx$
- ◆ For a constant k , $\int_a^b k \cdot f(x) dx = k \int_a^b f(x) dx$
- ◆ $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$
- ◆ $\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$

5. Suppose you know that $\int_2^5 f(t) dt = 7$. Do you have enough information to evaluate each of the following integrals? If so, evaluate; if not, explain why not.

(a) $\int_2^5 [f(t) + 4] dt$

Solution

Yes; we can rewrite this as

$$\begin{aligned} \int_2^5 [f(t) + 4] dt &= \int_2^5 f(t) dt + \int_2^5 4 dt \\ &= 7 + \int_2^5 4 dt \\ &= 7 + 12 \text{ by interpreting } \int_2^5 4 dt \text{ as a signed area} \\ &= \boxed{19} \end{aligned}$$

(b) $\int_2^5 f(t + 4) dt$

Solution

There is not enough information. The graph of $f(t + 4)$ is the graph of $f(t)$ shifted left by 4 units, so the behavior of $f(t + 4)$ on $[2, 5]$ has nothing to do with the behavior of $f(t)$ on $[2, 5]$.

(c) $\int_5^2 4f(t) dt$

Solution

Yes; we can rewrite this as

$$\begin{aligned}\int_5^2 4f(t) dt &= -\int_2^5 4f(t) dt \\ &= -4 \int_2^5 f(t) dt \\ &= -4 \cdot 7 \\ &= \boxed{-28}\end{aligned}$$

(d) $\int_2^4 [f(t) + t] dt + \int_4^5 f(t) dt$

Solution

Yes; we can rewrite this as

$$\begin{aligned}\int_2^4 [f(t) + t] dt + \int_4^5 f(t) dt &= \int_2^4 f(t) dt + \int_2^4 t dt + \int_4^5 f(t) dt \\ &= \int_2^5 f(t) dt + \int_2^4 t dt \\ &= 7 + \int_2^4 t dt \\ &= 7 + 6 \text{ (by visualizing } \int_2^4 t dt \text{ as a signed area)} \\ &= \boxed{13}\end{aligned}$$